Chapter 28
Martingales and Measures †

Chapter 29 prices a swaption by treating the annuity as a numeraire, Chapter 30 changes measure three times to derive three adjustments, Chapter 32 derives drift conditions from no-arbitrage, and Chapter 33 needs all of this at once. Reading this chapter first means none of those chapters has to borrow machinery it has not been given.

A reader who wants only the working rules can take Sections 28.1, 28.6 and 28.8 and move on. The rest is the justification. The vocabulary of Section 28.1 comes first because every practitioner uses it and few state it precisely: anyone who has traded a derivative has heard "risk-neutral probability" and "equivalent martingale measure," and comparatively few could say what either one is, though neither is hard.

The chapter runs from the vocabulary to what it buys. Sections 28.1 to 28.3 fix the terms: what a measure is, what the market price of risk is the price of, and what changes when uncertainty has more than one source. Sections 28.4 to 28.6 are the machinery: martingales, the change of measure written out explicitly, and the numeraire that names each measure. Sections 28.7 and 28.8 put it to work on why a forward price is an expectation and on the annuity measure from which Chapter 30's adjustment for constant-maturity swaps comes. Sections 28.9 to 28.11 restate the familiar results in this language, including what $N(d_1)$ is; Sections 28.12 and 28.13 are where a numeraire fails and what the theory is for.

28.1 THE VOCABULARY, STATED PRECISELY

Seven terms carry the whole subject. Each is stated here in the form a practitioner needs, which is less than a probabilist would want and more than the phrases usually get.

A probability measure P assigns a probability to each event, that is, each set of possible paths the market might follow. Two measures generally assign different probabilities to the same event, and need not even agree on which events can happen at all; the pairs that do agree on that are the subject of the next definition.

Equivalent measures. Two measures P and Q are **equivalent**, written $P \sim Q$, if they agree on which events have probability zero:

$$\text{for every event } A: \quad P(A) = 0 \iff Q(A) = 0. \tag{28.1}$$

They may disagree about how likely anything is, but not about what can happen. This is the precise sense in which a change of measure reweights the world without inventing or destroying scenarios, and it is exactly what a no-arbitrage argument needs, because an arbitrage is a payoff that is impossible to lose on, and impossibility must survive the change.

A self-financing portfolio is one whose value changes *only* because the prices of its holdings change: no cash is added and none is withdrawn after it is set up. Rebalancing is permitted, provided every purchase is paid for by a sale. The condition is what makes a replication argument an argument about prices, rather than about someone's willingness to keep funding a position, and it is why a payoff can be replicated at a cost that is known today. It is also the condition hidden in the next definition: an asset qualifies as a numeraire only if its quoted price

is the value of a self-financing position, which is why Section 28.12 has to disqualify a stock that pays dividends.

A numeraire is any traded asset with a strictly positive price, used as the unit in which other prices are quoted. Which one is chosen is not innocuous, and the freedom to choose is the technique that Part VI lives on: pick the numeraire that makes the payoff you are pricing simple, and the hard part of the calculation disappears into the choice.

A martingale is a process with zero drift: a process X is a martingale under Q if

$$E^Q[X(T) \mid \mathcal{F}_t] = X(t) \qquad \text{for all } t \le T, \tag{28.2}$$

where $\mathcal{F}_t$ is everything observable by time t, the **information** available then. The best forecast of a martingale's future value is its current value. A martingale is not a process that cannot move; it is one whose movements are unforecastable.

An equivalent martingale measure for a numeraire N is a measure $Q^N \sim P$ under which *every* traded asset price, divided by N, is a martingale:

$$\boxed{\frac{V(t)}{N(t)} = E^N\left[\frac{V(T)}{N(T)}\,\middle|\,\mathcal{F}_t\right]} \qquad \text{for every traded } V. \tag{28.3}$$

In a market free of arbitrage every numeraire has one, which is what makes the choice above a choice of measure. The word "equivalent" carries a condition and is usually skipped. It is what forbids the measure from assigning zero probability to a state the real world admits—without it, one could price away a genuine risk by declaring it impossible.

The risk-neutral measure is the equivalent martingale measure associated with one specific numeraire: the money market account $B(t)$, which grows at the instantaneous short rate. Under it,

$$V(0) = E^Q\left[e^{-\int_0^T r_s\,ds}\,V(T)\right],$$

and every asset has expected return r. The name misleads, so state what it is not. It is not a claim that investors are indifferent to risk, and the probabilities in it are not anyone's forecast. It is a pricing tool rather than a description of the world: a set of weights under which discounted prices have no drift, so that expectation becomes a valid pricing operation.

One sentence connects all seven. A price is a numeraire times an expectation, and the measure is whatever makes that true. Everything after this section proves that statement, draws a consequence from it, or chooses the numeraire well.

28.2 THE MARKET PRICE OF RISK

Start with a quantity that unifies everything after it. Suppose two assets depend on the same single source of uncertainty dW:

$$\frac{dg}{g} = \mu_g\,dt + \sigma_g\,dW, \qquad \frac{dh}{h} = \mu_h\,dt + \sigma_h\,dW$$

No-arbitrage forces the two to offer the same compensation per unit of risk:

$$\frac{\mu_g - r}{\sigma_g} = \frac{\mu_h - r}{\sigma_h} \tag{28.4}$$

The common value is the **market price of risk** λ carried by the source dW, and the proof that the two must agree is a portfolio.

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Example 28.1 What happens if they differ

Let $r = 3\%$, and take

$$\text{A} : \mu = 8\%,\ \sigma = 20\% \Rightarrow \lambda_A = 0.25, \qquad \text{B} : \mu = 6\%,\ \sigma = 15\% \Rightarrow \lambda_B = 0.20$$

Hold \$1 of A and −\$1.3333 of B, the ratio $-\sigma_A/\sigma_B$. The dW terms cancel exactly, so the portfolio is riskless. Its return is

$$1.0000 \times 8\% - 1.3333 \times 6\% = 0.00\%$$

and its cost is

$$1.0000 - 1.3333 = -0.3333$$

The portfolio has a negative cost (setting it up brings in \$0.3333 of cash) and a zero return. Investing that cash at 3% earns $0.03 \times 0.3333 = 0.0100$, so the position earns **1.00% of the notional of A, riskless, per year**.

Scale it and the profit is unbounded, which is what an arbitrage means. The requirement $\lambda_A = \lambda_B$ then pins B's expected return: at $\lambda = 0.25$ and $\sigma_B = 15\%$, μ_B must be $3\% + 0.25 \times 15\% = \mathbf{6.75}\%$.

This is asset pricing theory in one calculation, and the rest of the chapter restates it. Chapter 13's risk-neutral valuation is the case $\lambda = 0$; the measure changes of Sections 28.6 and 28.9 are the general case.

The market price of risk is a property of the risk, not of the asset. Two assets exposed to the same dW must carry the same λ whatever else differs about them: their maturity, their issuer, their payoff structure. In Chapter 31's one-factor models every bond carries the market price of interest rate risk, which is why a single λ suffices there and why the whole curve is determined by one process.

28.3 SEVERAL SOURCES OF UNCERTAINTY

Let $W_1, \dots, W_n$ be independent Wiener processes, and let the state of the economy be carried by n variables $\theta_1, \dots, \theta_n$ (**factors**, in the asset pricing literature) following

$$\frac{d\theta_i}{\theta_i} = m_i\,dt + s_i\,dW_i, \qquad i = 1, \dots, n$$

where m_i and s_i may depend on the factors and on time. The factors need not be traded, and usually are not: a short rate, an inflation index, a volatility, a default intensity. The force of the assumption is that nothing else moves: every price in this economy is a function of t and of those n numbers.

So let f be the price of a traded asset, $f(t) = F(t, \theta_1(t), \dots, \theta_n(t))$. Itô's lemma in n variables (Section 14.11 states the two-variable case) then delivers the dynamics rather than assuming them,

$$\frac{df}{f} = \mu\,dt + \sum_{i=1}^{n} \sigma_i\,dW_i, \qquad \sigma_i = \frac{\theta_i}{F}\frac{\partial F}{\partial \theta_i}\,s_i$$

where μ collects the time, drift and second-order terms, and no cross terms survive because the W_i are independent. Read σ_i as the asset's exposure to factor i: the elasticity of its price in that factor, times that factor's own volatility. Such an asset must satisfy

$$\mu - r = \sum_{i=1}^{n} \lambda_i \sigma_i \tag{28.5}$$

where λ_i is the market price of risk for dW_i. The claim being made is that the λ_i are the same numbers for every asset. Applied to one asset in isolation the equation restricts nothing; it merely names n quantities. Applied to all of them at once it is the whole of the theory.

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The proof is Example 28.1 with more bookkeeping. There, one asset hedged another and the riskless residual was forced to earn r. Here n assets hedge one.

Begin with n traded assets that are **non-redundant** (no one of them is a portfolio of the others) and whose exposures **span** the risks,

$$\frac{df_j}{f_j} = \mu_j\,dt + \sum_{i=1}^{n} \sigma_{ji}\,dW_i, \qquad j = 1, \ldots, n$$

spanning meaning that the matrix of loadings $[\sigma_{ji}]$ has rank n, hence is invertible: no risk is unhedgeable. Define $\lambda_1, \ldots, \lambda_n$ as the solution of

$$\mu_j - r = \sum_{i=1}^{n} \lambda_i\,\sigma_{ji}, \qquad j = 1, \ldots, n$$

n linear equations in n unknowns, solvable and uniquely so because the loadings are invertible. Nothing has been assumed at this point. The λ_i are simply being read off the market: they record what these n assets currently pay per unit of each risk.

Now take *any* other asset in the economy, with drift μ and exposures $\sigma_1, \ldots, \sigma_n$, and hedge it. Hold one dollar of it, and $-w_j$ dollars of each f_j, with the w_j chosen so that every exposure is matched:

$$\sum_{j=1}^{n} w_j\,\sigma_{ji} = \sigma_i, \qquad i = 1, \ldots, n$$

solvable, and uniquely, for the same reason as before. Every dW_i term now cancels and the portfolio is riskless. It costs $1 - \sum_j w_j$ to establish and drifts at $\mu - \sum_j w_j\mu_j$, so no-arbitrage requires the drift to be exactly r times the cost:

$$\mu - \sum_j w_j\,\mu_j = r\left(1 - \sum_j w_j\right)$$

Collecting the interest terms leaves a statement about excess returns alone,

$$\mu - r = \sum_j w_j\,(\mu_j - r)$$

and the remainder is substitution. Replace each $\mu_j - r$ by the defining relation above, exchange the order of summation, and the inner sum is the hedge condition:

$$\mu - r = \sum_j w_j \sum_i \lambda_i\,\sigma_{ji} = \sum_i \lambda_i \sum_j w_j\,\sigma_{ji} = \sum_i \lambda_i\,\sigma_i$$

which is (28.5).

Notice what the argument never did. The λ_i were pinned down by the n spanning assets and then never revisited, so they cannot depend on which asset was subsequently tested against them. The one-factor conclusion, that λ belongs to the risk and not to the asset, survives intact, and it survives for the same reason.

Suppose (28.5) fails. If some asset's drift exceeds $r + \sum_i \lambda_i\sigma_i$ by ε, the portfolio just built is riskless, costs nothing once the interest leg is financed, and earns ε per dollar per year. Reverse it if the drift falls short. Scale it and the profit is unbounded, so no asset can sit off the hyperplane (28.5) traces out.

Now count the assets against the risks. The argument used n non-redundant traded assets against n independent Wiener processes; with only $k < n$ of them non-redundant, the defining relations become k equations in n unknowns. Solutions still exist (no-arbitrage puts excess

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returns in the span of the loadings whatever the market's structure), but they form an $(n - k)$-dimensional family, every member of which prices the traded assets correctly, so no observed price singles one out. The rule is that *the market prices of risk are unique exactly when the traded assets are at least as numerous as the sources of uncertainty they must hedge*, counting only the non-redundant ones. Below that count, which is the incomplete-market case, one member of that family is chosen rather than read off the market; at or above it, every further asset must price consistently with the λ_i already pinned down, or be an arbitrage.

A market is complete (Section 13.4) when every payoff can be replicated by trading the assets it has. The **second fundamental theorem** of asset pricing says that an arbitrage-free market is complete exactly when its equivalent martingale measure is unique, so the arbitrariness of an incomplete market enters here, at the market price of risk. [TN28] states both theorems and proves them.

Two of the model families in this book have fewer traded assets than sources of risk. Chapter 21's stochastic volatility models add a second Wiener process driving the variance, and variance is not traded: two risks, one traded underlying, so the price of variance risk cannot be read off the market. It is chosen. Heston takes the variance risk premium to be λv, proportional to the variance, which keeps the variance in the same square-root class under Q, with Chapter 21's mean-reversion speed κ and long-run variance θ becoming $\kappa + \lambda$ and $\kappa\theta/(\kappa + \lambda)$. This is a modeling decision made for tractability, not a defect. It costs little: calibrating to option prices delivers the risk-neutral parameters directly, so the split between them and λ never has to be made, which is why the model is written under Q from the start. The premium itself does not disappear (it is Section 19.10's volatility risk premium), and once an option trades, its price supplies what the count could not. One traded option completes the market.

Chapter 31's short-rate models are the same case, and the same kind of choice is made there. The short rate drives everything and is not itself traded, so the price of rate risk is specified rather than observed: constant in Vasicek, proportional to $\sqrt{r}$ in Cox–Ingersoll–Ross, each form picked so that the process keeps its class under Q, Gaussian staying Gaussian and square root staying square root. Writing the model under Q from the start, as Chapter 31 does, is that choice already made, and calibrating to today's curve means the real-world parameters and λ never have to be separated.

The economic content of (28.5) is that risks price additively. An asset's excess return is the sum of its exposures times the prices of those exposures, which is the structure of every factor model in asset pricing, arrived at here from no-arbitrage rather than from equilibrium.

The same equation identifies what a model must supply. Chapter 32's K-factor Heath–Jarrow–Morton (HJM) model needs K market prices of risk to move from the real world to the risk-neutral one. Section 32.2's drift condition is exactly the statement that this move has been made, which is why the drift is not free.

28.4 MARTINGALES

Section 28.1 defined a martingale through its conditional expectation. In the continuous-time notation of Chapter 14 the same condition reads

$$dX = \sigma\, dW$$

a process with no dt term. Zero drift and (28.2) say the same thing under an integrability condition that holds throughout this book, and whose failure [TN28] describes.

The reason martingales matter is that they turn valuation into an expectation. If some ratio

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V/N is a martingale, then

$$\frac{V_0}{N_0} = E\left[\frac{V_T}{N_T}\right] \quad \Longrightarrow \quad V_0 = N_0\, E\left[\frac{V_T}{N_T}\right]$$

and the pricing problem is solved provided the expectation can be computed. The ratio is a martingale only when V and N are both self-financing, a condition Section 28.12 returns to. Which ratios are martingales, and under which probability measure, is the entire subject of this chapter.

28.5 THE CHANGE OF MEASURE, MADE EXPLICIT

Before the general theory, a two-period binomial tree shows a measure change as an actual list of numbers: in Figure 28.1 the nodes are fixed, the middle terminal node is reached two ways, and only the branch probabilities change with the measure.

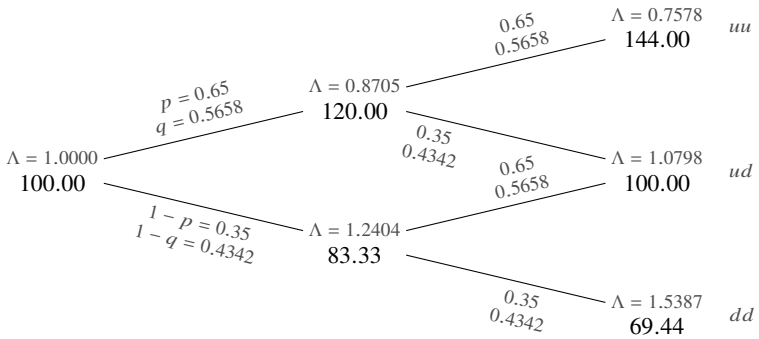


Figure 28.1: Example 28.2's tree: the real-world and risk-neutral probability on each branch, and Λ above the price at each node.

Example 28.2 The Radon–Nikodym derivative on a two-step tree

$S_0 = 100$, $u = 1.2$, $d = 1/1.2 = 0.8333$, $r = 4\%$, annual steps. The real-world probability of an up move is $p = 0.65$, an assumption about the world and one no one can observe directly.

The risk-neutral probability, Chapter 13's p, is not an assumption. It is forced:

$$q = \frac{e^{r\Delta t} - d}{u - d} = 0.565848$$

Both measures assign probabilities to the same three terminal nodes. The ratio of the two is the Radon–Nikodym derivative $\Lambda = dQ/dP$:

Node	$S(2)$	P	Q	Λ
dd	69.4444	0.122500	0.188488	1.538680
ud	100.0000	0.455000	0.491328	1.079842
uu	144.0000	0.422500	0.320183	0.757831
		1.000000	1.000000	$E^P[\Lambda] = 1.000000$

Three facts carry the theory. $E^P[\Lambda] = 1$ exactly, and every value is positive: no path P allows is impossible under Q, which is what equivalence means. Together these are what make Λ a legitimate change of probability rather than an arbitrary reweighting.

The reweighting is monotone in the outcome. Good states are weighted *down* (0.7578) and bad states *up* (1.5387). The direction is not automatic: it holds because $p > q$, which is this tree's way of saying the stock carries a risk premium, $e^{-2r}E^P[S(2)] = 106.02$ against $S_0 = 100$. That is what the market price of risk means, expressed as probabilities: a dollar delivered in a bad state is

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worth more to an investor than a dollar delivered in a good one, so a valuation that discounts every state at r has to make up the difference by treating bad states as more likely than they are. Were there no premium ($p = q$), the two measures would coincide, and a negative premium would tilt dQ/dP the other way.

And the expectation comes out right: $e^{-2r}E^Q[S(2)] = 100.0000 = S_0$, so the discounted stock price is a Q-martingale.

Now price a two-year call struck at 100:

$e^{-2r}\,E^P[\text{payoff}]$	17.160733
$e^{-2r}\,E^Q[\text{payoff}]$	13.004930
Error from using the wrong measure	**+32.0**%

Same nodes, same payoffs—only the probabilities differ, and the price is wrong by a third.

In general the object has a name, and a theorem behind it. Equivalence says P and Q agree on which events are impossible, so neither puts weight where the other puts none; the **Radon–Nikodym theorem** says that this is enough for Q to be P reweighted by a single random variable. That variable is the **Radon–Nikodym derivative** $\Lambda = dQ/dP$, also called the **density** of Q with respect to P, positive, with

$$Q(A) = E^P[\Lambda\,\mathbf{1}_A] \quad \text{for every event } A, \qquad E^P[\Lambda] = 1. \tag{28.6}$$

The unit mean is forced by taking A to be everything, and expectations follow the same rule:

$$E^Q[X] = E^P[\Lambda X] \quad \text{for any payoff } X. \tag{28.7}$$

Changing measure is multiplying by Λ before averaging, never a change in which outcomes are possible. On a tree Λ is a list, one number per path (the list above, where ud and du share a value); in continuous time it is a process, $\Lambda_t = E^P[\Lambda \mid \mathcal{F}_t]$, and Section 28.6 identifies which one. That conditional expectation is concrete. Λ is known only at the end, once the path is complete; Λ_t is its P-average over the paths still open at t.

At $t = 0$ nothing is ruled out, so $\Lambda_0 = E^P[\Lambda] = 1$; at T one path is left and $\Lambda_T = \Lambda$. At the upper $t = 1$ node the open paths are uu and ud, so $\Lambda_1 = 0.65 \times 0.757831 + 0.35 \times 1.079842 = 0.870535$, which is q/p; the lower node gives $(1 - q)/(1 - p) = 1.240436$. On a tree the density is the running product of one-step ratios. It is a P-martingale by the **law of iterated expectations** (also called the tower property), which says that averaging a later conditional expectation over what is known earlier gives the earlier one:

$$E\big[E[X \mid \mathcal{F}_s] \,\big|\, \mathcal{F}_t\big] = E[X \mid \mathcal{F}_t], \qquad t \le s.$$

Here $0.65 \times 0.870535 + 0.35 \times 1.240436 = 1$ again. The density also reweights any event settled by t: $Q(A) = E^P[\Lambda_t\,\mathbf{1}_A]$ for A in $\mathcal{F}_t$.

28.6 NUMERAIRES AND THE MEASURES THAT GO WITH THEM

The martingales of this chapter are prices measured in units of a numeraire, and Section 28.1 has already stated the central result: in a market free of arbitrage every numeraire N has the measure Q^N of (28.3) ([TN28] gives the continuous-time conditions). What matters is not that the measures exist but how to move between them, and that is one line.

> **The change-of-numeraire theorem.** Let N and M be two numeraires with associated measures Q^N and Q^M. Their density, restricted to the information $\mathcal{F}_t$ available at time t (the bar is read that way), is
>
> $$\left.\frac{dQ^M}{dQ^N}\right|_{\mathcal{F}_t} = \frac{M(t)\,/\,M(0)}{N(t)\,/\,N(0)}. \tag{28.8}$$

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The left side is the density process Λ_t of Section 28.5, written for this pair of measures. The theorem restates the definition rather than adding a fact. Define a measure by taking (28.8) as its density with respect to Q^N, and compute the expectation of $V(t)/M(t)$ under it for any traded V, by (28.7):

$$E^M\left[\frac{V(t)}{M(t)}\right] = E^N\left[\frac{V(t)}{M(t)}\cdot\frac{M(t)/M(0)}{N(t)/N(0)}\right] = \frac{N(0)}{M(0)}\,E^N\left[\frac{V(t)}{N(t)}\right] = \frac{N(0)}{M(0)}\cdot\frac{V(0)}{N(0)},$$

which is $V(0)/M(0)$. Conditioned on $\mathcal{F}_s$, with s in place of 0, the same lines give $V(s)/M(s)$, so the measure is Q^M. The ratio in (28.8) is positive, both numeraires being strictly positive, and it is a Q^N-martingale with unit expectation, so it is a valid density.

One specialization generates most of Part VI. Taking $N = P(\cdot, T_2)$ and $M = P(\cdot, T)$, two zero-coupon bonds maturing in either order, for $t \le \min(T, T_2)$,

$$\left.\frac{dQ^T}{dQ^{T_2}}\right|_{\mathcal{F}_t} = \frac{P(t,T)\,/\,P(0,T)}{P(t,T_2)\,/\,P(0,T_2)}, \tag{28.9}$$

and (28.7) gives, for any X measurable at t,

$$E^T[X] = E^{T_2}\left[X\cdot\frac{dQ^T}{dQ^{T_2}}\right] = E^{T_2}[X] + \mathrm{Cov}^{T_2}\left(X, \frac{dQ^T}{dQ^{T_2}}\right), \tag{28.10}$$

since the density has unit mean. Every adjustment in Chapter 30 is an application of (28.10), with the relevant pair of numeraires in place of the two bonds. And the condition for no adjustment falls straight out: if X is uncorrelated with the ratio of the two bond prices, the covariance vanishes and the two expectations agree. That is why a convexity, timing, or quanto correction is always a covariance with the numeraire ratio, which moves with the discount rate in a convexity or timing correction and with the exchange rate in a quanto. Four numeraires account for essentially all of practice; the table lists what each is for.

Numeraire	Measure	What becomes a martingale	Used for
Money market	Risk-neutral	V/B, $B(t) = \exp\left(\int_0^t r_s\,ds\right)$	Simulation
Zero-coupon bond $P(t,T)$	T-forward	Any T-maturity forward	Single payoffs
Swap annuity $A(t)$	Swap	The forward swap rate	Swaptions, CMS
Foreign money market	Foreign risk-neutral	Foreign assets	Quantos, FX

The foreign money market prices a payoff fixed in foreign currency: under it a foreign asset drifts at the foreign rate, and a quanto differs from it by the asset's covariance with the exchange rate, Section 30.6's quanto adjustment. Two remarks are easy to miss.

The choice of numeraire does not change any price. That is the content of the theorem, and Section 31.7 verifies it numerically. But it changes the difficulty enormously, and that is the practical point. A quantity that requires a simulation under one numeraire may have a closed form under another.

28.7 WHY THE FORWARD PRICE IS AN EXPECTATION

Chapter 5 derived the forward price from arbitrage, in the form $F(t,T) = S_t/P(t,T)$ that holds whatever rates do; Chapter 30 will open by asserting that a forward price is an expectation under the right measure. This section makes "the right measure" precise, and shows what goes wrong under the wrong one.

Under the T-forward measure (numeraire $P(t,T)$), the forward price $S_t/P(t,T)$, a traded asset divided by the numeraire, is a martingale. For a stock,

$$\frac{S_0}{P(0,T)} = E^T\left[\frac{S_T}{P(T,T)}\right] = E^T[S_T]$$

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since $P(T,T) = 1$. So $F(0,T) = E^T[S_T]$, exactly, which is the result Chapter 30 uses throughout.

The futures price, by contrast, is a martingale under the risk-neutral measure. The proof is a trading strategy. The money market account $B(t)$ follows

$$dB = rB\,dt$$

Write $H(t)$ for the futures price for delivery at T. A futures contract costs nothing to enter and pays dH in cash as it accrues, so consider holding $B(t)$ contracts at every t and keeping the proceeds in the account. The strategy starts from nothing, and its value $V(t)$ is the balance in the account, earning interest r. Since a futures position is itself worth zero once it has been marked to market, the change in V can be written as

$$dV = \underbrace{rV\,dt}_{\text{interest on the balance}} + \underbrace{B\,dH}_{\text{settlement on } B(t) \text{ contracts}}$$

Those are the only cash flows there are, which is what makes the strategy self-financing. Dividing through by the account,

$$d\left(\frac{V}{B}\right) = dH, \qquad \text{so} \qquad \frac{V(t)}{B(t)} = H(t) - H(0)$$

Section 28.6 makes V/B a Q-martingale, because V is the value of a self-financing portfolio of traded assets; so the futures price H is a Q-martingale too. At delivery $H(T) = S_T$, which gives the pair

$$\text{futures price} = E^Q[S_T], \qquad \text{forward price} = E^T[S_T] \tag{28.11}$$

Two prices for the same delivery, two measures, and the difference between them is the futures–forward convexity adjustment of Section 30.3. Section 5.12 reaches it without the machinery, by asking when the variation margin arrives.

28.8 THE ANNUITY MEASURE, AND WHERE CMS COMES FROM

The third numeraire in the table is the least intuitive and does the most work. A swap that starts at T_0 pays its fixed leg on dates $T_1, \ldots, T_N$, and the annuity

$$A(t) = \sum_{i=1}^{N} \tau_i\, P(t, T_i)$$

is the present value of the fixed leg per unit of fixed rate on unit notional, τ_i being the year fraction of the ith period. It is a portfolio of zero-coupon bonds, hence a traded asset, hence an admissible numeraire.

The forward swap rate is

$$S(t) = \frac{P(t,T_0) - P(t,T_N)}{A(t)}$$

a traded asset divided by the annuity, so it is a martingale under the annuity measure Q^A. That is the justification for Section 29.6.

Now notice what a swaption's payoff looks like. It is $A(T_0)\max(S(T_0) - K, 0)$: the annuity appears in the payoff. Valuing it with A as numeraire,

$$V_0 = A(0)\,E^A[\max(S(T_0) - K, 0)]$$

and the annuity cancels. With S a martingale and lognormal, Black's formula follows exactly, which is why the swaption convention works as well as it does.

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That cancellation is this chapter's thesis in its simplest form. Nothing was approximated and no model of the curve was needed: a numeraire was picked so that the awkward part of the payoff divided out. Most of Part VI consists of cases where the cancellation is imperfect, and what fails to cancel is precisely what has to be computed.

The nearest such case differs from a swaption only in how the rate is paid, and it recurs throughout Part VI. In a constant-maturity swap, one leg pays a long swap rate (the ten-year rate, say) observed afresh on each reset date, against a fixed rate or a short floating rate on the other leg. The floating leg of an ordinary swap references a rate whose own life matches the period it accrues over: the three-month rate set in January covers January to April, and is paid at the end of it. A CMS leg breaks that alignment deliberately. The ten-year rate set in January is paid three months later, on a quarterly accrual, and the ten-year swap it is the rate *of* never happens.

Investors buy the leg for the exposure it isolates: a coupon that follows long rates without the maturity, the duration, or the balance-sheet consequences of holding a ten-year bond, and dealers hedge the resulting book in swaptions. It is also the building block of steepener notes, which pay a multiple of a CMS spread, the difference between two CMS rates of different tenors, such as the ten-year rate less the two-year.

The coupon pays $S(T_0)$ on a *single* date T_p rather than over the annuity, so the numeraire it calls for is the zero-coupon bond $P(\cdot, T_p)$ rather than A. Under that measure S is no longer a martingale, the annuity no longer cancels, and what is left behind is the covariance between S and the ratio of the two numeraires.

That covariance is the CMS convexity adjustment, which Section 30.5 computes, sizes, and draws. What belongs here is where it comes from: it is not a correction for a modeling deficiency and not an empirical patch, but the price of quoting a rate that is a martingale under one measure in a contract that settles under another.

Even its sign follows from the cancellation that failed. A swap rate that rises raises the coupon linearly but lowers the value of the annuity the coupon would have been paid over, and the CMS contract keeps the first effect while discarding the second—so the adjustment is positive. The holder of a CMS coupon has been handed that convexity, and the adjustment is what it costs.

28.9 GIRSANOV

Changing the measure changes the drift and leaves the volatility alone. Formally: let dW^N be a Wiener process under Q^N with correlation ρ to the numeraire ratio M/N, whose volatility is $\sigma_{M/N}$. Moving to Q^M,

$$dW^M = dW^N - \rho\, \sigma_{M/N}\, dt$$

is a Wiener process under Q^M. It is a new process and not the old one relabeled (the two differ by that drift), and only the part of dW^N that moves with the numeraire is shifted, which is where ρ comes from. Every process therefore picks up a drift equal to the covariance between its own logarithm and the log of the numeraire ratio, and a process uncorrelated with that ratio keeps the drift it had. That single sentence generates Chapter 30's three adjustments and Chapter 32's two drift conditions, each derived where it is used.

The simplest case shows the whole mechanism. Take two assets following geometric Brownian motions under the risk-neutral measure, with correlation ρ between their Wiener processes:

$$\frac{dS_1}{S_1} = r\, dt + \sigma_1\, dW_1, \qquad \frac{dS_2}{S_2} = r\, dt + \sigma_2\, dW_2$$

Now elect S_2 as the numeraire. The numeraire ratio is S_2/B, whose volatility is σ_2, so the rule above adds to S_1's drift its covariance with $\ln(S_2/B)$, which is $\rho\sigma_1\sigma_2$, and to S_2's its own

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variance σ_2^2:

$$\frac{dS_1}{S_1} = (r + \rho\sigma_1\sigma_2)\,dt + \sigma_1\,dW_1^{S_2}, \qquad \frac{dS_2}{S_2} = \left(r + \sigma_2^2\right)dt + \sigma_2\,dW_2^{S_2}$$

The two Wiener processes are shifted by different amounts, $dW_1^{S_2} = dW_1 - \rho\sigma_2\,dt$ and $dW_2^{S_2} = dW_2 - \sigma_2\,dt$, because S_2 moves with itself perfectly and with S_1 only through ρ.

Both drifts moved and neither volatility did. σ_1 and σ_2 multiply the new Wiener processes unchanged; what changed is the rate at which each asset is expected to grow, and by exactly the covariance the rule names.

The check is that the ratio is now a martingale. Itô's lemma applied to $G = S_1/S_2$, as in Example 14.7, gives it a drift of

$$(r + \rho\sigma_1\sigma_2) - \left(r + \sigma_2^2\right) + \sigma_2^2 - \rho\sigma_1\sigma_2 = 0$$

as it must have, since G is a traded asset divided by the numeraire. Its volatility is $\sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$, so the change of numeraire has already produced the volatility that Section 28.11 prices an exchange option on, before any option has been mentioned.

A change of measure reweights the probability of paths. It can make an upward path more likely and a downward path less likely, which moves the mean. But it cannot change how far apart the paths are, because the paths themselves are unchanged—only their probabilities move. Two consequences follow directly and are used constantly.

Volatility is the same in every measure, so a volatility estimated from real-world data can be used in a risk-neutral pricing model, and historical and implied volatility can be compared at all. What is invariant is the volatility of a path. Where volatility is itself random, its expected future level can still differ between measures, and that difference is Section 19.10's volatility risk premium.

Expected returns are not. A traded asset drifts at μ under P and at r under Q, and the change of measure shifts dW by the same $\lambda\,dt$ for every asset it drives, so $\mu - r = \lambda\sigma$ with one λ for all: the market price of risk of (28.4), reached here by a change of measure rather than by Example 28.1's hedge.

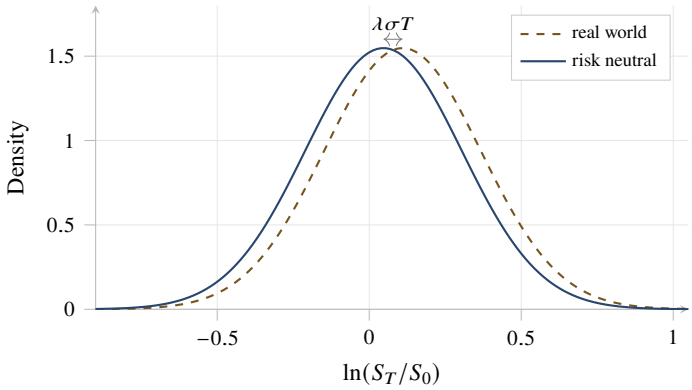


Figure 28.2: Example 28.2's tree in continuous time, at its two-year horizon: one density displaced by $\lambda\sigma T$.

Figure 28.2 draws the theorem on Example 28.2's tree, whose steps imply $\sigma = \ln 1.2 = 18.23\%$ and whose p implies a drift $\mu = 7.13\%$: the same density drawn twice, displaced by $(\mu - r)T = \lambda\sigma T = 0.0626$ and not otherwise altered. What is invariant is the volatility in the pathwise sense, the quadratic variation $\sigma^2 T$ of $\ln S$, which equivalent measures agree on path

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by path; that is a continuous-time fact, not a general one, and reweighting the two-step tree does change the variance, mildly. The two densities share the standard deviation $\sigma\sqrt{T}$ only because λ is constant here: a market price of risk that moved with the path would shift the mean by a random amount and change the spread as well. In the level of S_T the two give lognormals of visibly different shape, which is why the figure is drawn in logs.

28.10 WHAT $N(d_1)$ IS

Apart from Section 28.9's illustration, every numeraire so far has been a money market account, a bond, or a portfolio of bonds. Nothing in Section 28.6 required that, and dropping the restriction explains a formula the reader has known since Chapter 15.

Split the Black–Scholes call into its two pieces:

$$c = e^{-rT}E^{Q}\left[S_T\mathbf{1}_{\{S_T>K\}}\right] \;-\; Ke^{-rT}\,Q(S_T > K) \tag{28.12}$$

The second piece is the easy one. $Q(S_T > K)$ is a probability under a measure in which $\ln S_T$ is normal with mean $\ln S_0 + (r - \sigma^2/2)T$, and it evaluates to $N(d_2)$. So $N(d_2)$ is the risk-neutral probability that the option finishes in the money, which is the one part of the formula most readers can already interpret.

The first piece is the awkward one, because it is the expectation of a *product*: a terminal price multiplied by an indicator of an event that depends on that same price. Done directly it means integrating S_T against a lognormal density over a half-line and completing a square.

Choose the numeraire to make the product disappear. The payoff $S_T\mathbf{1}_{\{S_T>K\}}$ is the promise to deliver one share if the share finishes above K. Value it using the *stock itself* as the numeraire, which is admissible since a non-dividend-paying stock is traded and strictly positive:

$$\frac{V_0}{S_0} = E^{S}\left[\frac{S_T\mathbf{1}_{\{S_T>K\}}}{S_T}\right] = E^{S}\left[\mathbf{1}_{\{S_T>K\}}\right] = Q^{S}(S_T > K)$$

Dividing by S_T is exactly what cancels the S_T in the payoff, and an expectation of a product has become a probability. So $V_0 = S_0\,Q^{S}(S_T > K)$.

What remains is that probability, and Girsanov supplies it. Moving from the money-market numeraire to the stock shifts every drift by its covariance with $\ln S$; for the stock itself that covariance is σ^2, so its log-drift under Q^S is $r + \sigma^2/2$ rather than the $r - \sigma^2/2$ it has under Q. The variance is untouched, and the same calculation that produced $N(d_2)$ now produces

$$Q^{S}(S_T > K) = N(d_1), \qquad d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

Substituting both pieces into (28.12) gives $c = S_0N(d_1) - Ke^{-rT}N(d_2)$.

This says something about the Black–Scholes–Merton formula that the usual derivation hides. $N(d_1)$ and $N(d_2)$ are the probability of *the same event*, the option finishing in the money, computed under *two different measures*. The formula is one event priced in two units of account: shares for the first term, cash for the second. Calling $N(d_1)$ "the delta" is true and unilluminating; it is a probability, and it differs from $N(d_2)$ only because it is measured in a different numeraire. Example 28.3 puts numbers on the gap.

Example 28.3 The same event, two probabilities

Take a one-and-a-half-year call with $S_0 = 100$, $K = 110$, $r = 4\%$, $\sigma = 28\%$. Then

$$d_1 = \frac{\ln(100/110) + \left(0.04 + 0.28^2/2\right)\times 1.5}{0.28\sqrt{1.5}} = 0.068498, \qquad d_2 = 0.068498 - 0.28\sqrt{1.5} = -0.274431$$

so $N(d_1) = 0.527305$ and $N(d_2) = 0.391877$, and the call is worth $100\times 0.527305 - 110e^{-0.06}\times 0.391877 = \mathbf{12.1344}$.

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The option is out of the money and the two probabilities disagree by 13.5 percentage points. Under the risk-neutral measure it finishes in the money **39.19%** of the time; under the stock measure, **52.73%**—more likely than not.

Neither number is a forecast, and the gap is not an error. The stock measure over-weights the paths that finish high, because it measures value in shares and shares are worth more precisely on those paths. That reweighting is the whole content of d_1 versus d_2, and the distance $\sigma\sqrt{T} = 0.3429$ between d_1 and d_2 is the standard deviation of the log numeraire ratio over the option's life.

28.11 EXCHANGING ONE ASSET FOR ANOTHER

The technique is at its most persuasive on a payoff that looks genuinely harder. An **exchange option** confers the right to give up asset 2 and receive asset 1:

$$\max(S_1(T) - S_2(T),\, 0)$$

The payoff has two assets, two volatilities, a correlation, and *a strike that moves*. Attacked directly this is a two-dimensional problem: a partial differential equation in two state variables, or a double integral over a bivariate lognormal.

Use asset 2 as the numeraire and the second dimension disappears. Asset 2 is traded and strictly positive, so it is admissible, and

$$\frac{V_0}{S_2(0)} = E^{S_2}\left[\frac{\max(S_1(T) - S_2(T),\, 0)}{S_2(T)}\right] = E^{S_2}\left[\max(G_T - 1,\, 0)\right], \qquad G = \frac{S_1}{S_2}$$

The moving strike has become the number 1, which is what electing an asset as the unit of account buys. And G is a traded asset divided by the numeraire, so it is a martingale under Q^{S_2}, with no drift to estimate.

The volatility of G is the ratio volatility of Example 14.7:

$$\sigma_G = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2} \tag{28.13}$$

A driftless lognormal struck at 1 is Black's formula, and unwinding the numeraire gives **Margrabe's formula**:

$$V_0 = S_1(0)\,N(d_1) - S_2(0)\,N(d_2) \tag{28.14}$$

where

$$d_1 = \frac{\ln(S_1(0)/S_2(0)) + \sigma_G^2 T/2}{\sigma_G\sqrt{T}}, \qquad d_2 = d_1 - \sigma_G\sqrt{T}$$

The result has four features that matter. The interest rate is absent. In units of asset 2 there is nothing to finance, so no rate can appear.

The Black–Scholes–Merton formula is the special case. Let asset 2 be K zero-coupon bonds maturing at T, so $S_2(0) = Ke^{-rT}$ and $\sigma_2 = 0$. Then (28.13) collapses to σ_1 and (28.14) reads

$$V_0 = S_1(0)\,N(d_1) - Ke^{-rT}\,N(d_2)$$

An ordinary call is an option to exchange cash for stock, and the two formulas are one formula.

The reading of $N(d_1)$ and $N(d_2)$ from Section 28.10 survives the generalization.

$$N(d_2) = Q^{S_2}(S_1(T) > S_2(T)), \qquad N(d_1) = Q^{S_1}(S_1(T) > S_2(T))$$

It is one event, priced first in units of asset 2 and then in units of asset 1. Which asset is the numeraire is a choice, and the formula is symmetric in a way the Black–Scholes special case conceals.

And the correlation is what the value turns on. With $S_1(0) = S_2(0) = 100$, $\sigma_1 = 30\%$, $\sigma_2 = 20\%$ and one year, (28.13) gives $\sigma_G = 26.46\%$ at $\rho = 0.5$ and the option is worth 10.52;

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at $\rho = 0$ it is worth 14.31, and at $\rho = -0.5$, 17.25. Example 34.8 runs a similar pair from -0.5 to $+0.99$ and the value falls sevenfold: the input a desk can observe, the interest rate, does not affect the price, and the input it cannot observe, the correlation, determines it.

Exchange options are rarely labeled as such. An asset manager paid on performance against a benchmark holds one, since the fee is a call on the fund's value with the index as the strike. So is the quality option in a Treasury futures contract: the short delivers whichever bond is cheapest, and the switch between two candidates (Section 6.4) is the right to deliver one bond in place of the other. A gas-fired power station is a strip of them (Section 36.6's spark spread option is the option to exchange fuel for electricity, exercised hour by hour). Chapter 34 develops the multi-asset family and Chapter 36 the energy applications; all of them are one line of change of numeraire.

28.12 WHEN A NUMERAIRE FAILS

Every change of numeraire in this chapter has assumed two things about the asset in the denominator: that it is **traded**, meaning its price is the value of a self-financing portfolio, and that its price is **strictly positive**. Both assumptions fail in practice. The four cases below run in increasing order of what they cost. Each of the first three violates one of the two conditions; the fourth violates neither, and is the most consequential of all.

1. The price is not a self-financing portfolio

A stock paying dividends is not a numeraire, because the holder receives cash along the way, so its price is not the value of a self-financing position. The repair is complete and costs nothing: use the *reinvested* stock instead. Chapter 5's dividend adjustments are that same repair.

2. The price can reach zero

Dividing by a numeraire that can vanish is not permitted, and defaultable assets vanish. An option on a credit default swap takes as its numeraire the forward risky annuity of Section 25.6 (the present value of one basis point paid from expiration until default or maturity), under which the forward CDS spread is a martingale. But that annuity is worth nothing in default, so the measure it defines gives probability zero to an event that Q does not, and the two are therefore not equivalent in the sense (28.1) requires.

Single-name convention steps around this by making the option knock out at default. Index options do not knock out: the protection buyer is owed the losses accumulated before expiration, which is what front-end protection (Section 25.6) is, so the correction matters most in precisely the state that destroys the numeraire. Morini and Brigo (2011) construct the conditional measure that repairs it. Negative rates are not an instance of this failure, despite the association (a zero-coupon bond is worth *more* than par when rates are below zero, and stays positive), and what broke in the euro and yen swaption markets after 2014 was the lognormal quoting convention, repaired by Section 29.8's shifted and normal conventions.

3. The numeraire is not an asset at all

This is the quiet failure. The par-yield cash-settlement convention, the euro market's standard until 2018, values the contract with

$$\tilde{A}(S) = \sum_{i=1}^{mn} m^{-1}(1 + S/m)^{-i}$$

for an n-year swap paying m times a year, a formula applied to the swap rate rather than a portfolio of bonds, so it is not a traded asset, and it does not cancel against the numeraire the way Section 28.8 required. The cash annuity falls as rates rise: on a ten-year annual-pay swap,

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from 8.98 at a 2% swap rate to 7.36 at 6%, so it is smallest exactly where a payer swaption pays most, and that covariance survives. On an at-the-money five-year option on a ten-year swap, at a 3% forward rate and 25% volatility, it takes about 9.9 basis points off a 66-basis-point premium (both quoted as a rate, per unit of annuity), so the cash-settled payer is 15% cheaper, widening to 28% at a volatility of 40% because it is a convexity term. Since November 2018 ISDA's default for euro swaptions has been a collateralized cash price computed from the clearing houses' discount curve, which removes the mismatch; the par-yield convention survives in older trades and wherever the parties elect it.

The contract pays $\tilde{A}(S_T)(S_T - K)^+$ in cash at expiration T, so with the true annuity as numeraire its value is

$$V_0 = A_0\, E^A\left[\frac{\tilde{A}(S_T)}{A_T}\,(S_T - K)^+\right] \tag{28.15}$$

and the ratio inside is the whole difficulty: A_T is a portfolio of bonds depending on the entire curve at expiration, $\tilde{A}(S_T)$ a formula in one rate, and nothing cancels. As written, the expectation runs over the *joint* law of the swap rate and the curve around it, which no swaption smile supplies.

One step removes the joint law, and it approximates nothing. The payoff depends on S_T alone, so conditioning on S_T and applying the law of iterated expectations gathers every unknown into a single function of the swap rate:

$$V_0 = A_0\, E^A\big[\alpha(S_T)\,\tilde{A}(S_T)(S_T - K)^+\big], \qquad \alpha(s) \equiv E^A\left[\frac{1}{A_T}\,\middle|\, S_T = s\right]. \tag{28.16}$$

What remains is a one-dimensional integral against the *marginal* law of S_T, which a smile does supply, with everything unknown pushed into α, the **annuity mapping function**. Nor is α arbitrary: a unit payoff at expiration must be worth the discount bond, forcing $A_0 E^A[\alpha(S_T)] = P(0,T)$. Freezing the annuity at its forward level $A_0/P(0,T)$ makes α the constant $P(0,T)/A_0$, which satisfies it on any curve; on a flat curve that constant is $1/\tilde{A}(S_0)$, which yields the figure quoted above; desks run linear terminal-swap-rate models instead, taking α affine in s, and the difference shows in the wings.

A physically settled swaption has a price once the volatility is given. A cash-settled one has a price once the volatility *and* a model of the annuity are given—and a convention that reads as a settlement detail has moved the contract from the first category into the second.

4. Both conditions hold and the numeraire is still wrong

The risk-free bank account is traded and strictly positive, and it is not the numeraire for a collateralized trade. Nothing in the theorem flags the error: the wrong admissible asset has been chosen, and that is why this case is the expensive one. It closes a loop opened in Part I.

Chapter 7 argued that a collateral agreement determines the discount curve, and Chapter 8 that the terms of the CSA change the value of a trade. In this chapter's language, those are statements about which numeraire is correct. A trade collateralized in cash earning the overnight rate is replicated by a position financed at that rate, so the correct numeraire is the *collateral account*, not a hypothetical risk-free bank account.

Three chapters' worth of apparatus follows from that one identification. *Multi-curve discounting* is the observation that the numeraire changed and the forecast curve did not. *ColVA* of Section 9.6 is the value of the optionality in *which* numeraire applies, since a CSA permitting several collateral currencies gives the poster a choice, and a choice of numeraire that one party controls has value. And *FVA* of Section 9.4 is what appears when the trade is *not* collateralized, so that no traded numeraire replicates the financing, which is why the FVA debate was, in part, a disagreement about whether a firm's own funding cost can define a numeraire, and why it

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could not be settled by better data. The question "what is the numeraire" is not abstract; it is the question of what portfolio actually replicates the trade, and each time the market's answer changed, the value moved with it: when CSAs made the collateral rate the discount rate after 2008, and when CME and LCH moved dollar cleared swaps to SOFR discounting on October 16, 2020 (Chapter 7). Where no collateral is posted, the firm's own funding stands in, and that is FVA.

28.13 WHAT THE THEORY IS FOR

A fair question, at the start of a part that will use this material on every page, is what it buys. It supplies the drift conditions. Chapter 32's HJM condition and LIBOR market model (LMM) drifts are Girsanov applied twice. Without the theory they are unmotivated formulas, and a desk cannot validate an unmotivated formula.

It makes Black's formulas exact. Section 29.3's caplet and Section 29.6's swaption are not approximations under their own measures: given a lognormal rate they follow exactly from the martingale property, and knowing which measure makes each exact tells you when to trust them.

It generates Chapter 30's adjustments. Convexity, timing, and quanto adjustments are all the same computation: a rate is a martingale under one measure and the payoff lives under another, so a covariance term appears.

It explains where models are needed and where they are not. The second fundamental theorem (Section 28.3) says in advance which prices will depend on a modeling choice, a predictive statement rather than a retrospective one. [TN28] works out the range of prices that incompleteness leaves open.

And it disciplines the use of probabilities. The measure that prices correctly is not the one that forecasts correctly—Section 28.1's distinction, missed often enough to matter.

What it does not do is tell you which model to use. The theory says a measure exists; it does not say what the volatility structure is or how many factors the curve has. Chapters 31 and 32 answer those questions by calibration and by calculation; the theory cannot.

SUMMARY

The market price of risk $\lambda = (\mu - r)/\sigma$ must be the same for any two assets driven by the same uncertainty; if it is not, a portfolio of the two is riskless and earns more than r. With several sources, $\mu - r = \sum_i \lambda_i \sigma_i$: risks price additively, derived from no-arbitrage rather than equilibrium. Risk-neutral valuation is the case $\lambda = 0$.

A change of measure reweights paths without changing which paths are possible. It moves drifts and leaves volatilities alone, which is why a volatility estimated from history can be used in a risk-neutral model at all, and the error from pricing under the wrong measure is not small: even on a two-period tree, real-world probabilities misprice a call substantially.

Valuation is the search for a ratio that is a martingale. For any numeraire (any traded, strictly positive, self-financing asset) there is a measure under which every price divided by it is driftless, so $V_0 = N_0 E[V_T/N_T]$. The choice changes no price and changes the difficulty enormously.

Each numeraire is chosen for what it cancels. A zero-coupon bond makes the forward rate a martingale, so Black's caplet formula is exact for a lognormal forward, and the forward price is an expectation under that measure, which in general it is not under the risk-neutral one. The annuity does the same for the swap rate, and because a swaption's payoff contains the annuity it cancels; a CMS coupon pays on a single date, the cancellation fails, and what survives is the convexity adjustment. The stock itself turns $E^Q[S_T \mathbf{1}_{\{S_T > K\}}]$ into a probability, which is what $N(d_1)$ is, and a second risky asset prices the right to exchange one asset for another, in which

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no interest rate appears at all. The gap between the forward and the futures price $E^{Q}[S_T]$, Section 30.3's futures–forward adjustment, is one such covariance term, and Chapter 31 shows another: the error from discounting a caplet at today's zero rate.

A numeraire can fail in practice, and the costlier failures are the less visible ones: the cash-settled swaption annuity is not a traded asset at all, so nothing cancels, and the risk-free bank account, which meets both conditions, is not the numeraire for a collateralized trade, which is what Chapter 7's discount curve and Chapter 9's ColVA and FVA are all statements about.

A market is arbitrage-free if and only if an equivalent martingale measure exists, and complete if and only if that measure is unique; [TN28] proves both and says what has to change in continuous time. The second theorem is where every difficulty in this book lives: when a price depends on a modeling choice, it is because the market is incomplete in a way that can be named.

FURTHER READING

Harrison and Kreps, and Harrison and Pliska, are the papers that established the two fundamental theorems; they are demanding, and are where to go for the results in full generality. Geman, El Karoui, and Rochet's paper on change of numeraire is the source for Section 28.6. Baxter and Rennie's short book gives the ideas without measure-theoretic machinery, and Shreve's two volumes are the standard graduate treatment for someone who wants the machinery. Björk's book sits between them.

PRACTICE QUESTIONS

28.1 Using Example 28.1, construct the riskless portfolio and compute its excess return, then explain why the market price of risk must be the same for two assets driven by the same source of uncertainty.

28.2 Using Example 28.2, state the three values of $\Lambda = dQ/dP$, explain why they average to one under P rather than under Q, and explain why the reweighting is monotone in the terminal price.

28.3 Using Example 28.2, state the size of the pricing error from using real-world probabilities and explain why it is positive for a call.

28.4 Explain what it means for a ratio to be a martingale, and why finding one is the whole of the valuation procedure.

28.5 Name the four numeraires that cover practice, state the measure each induces, and explain why the choice changes no price but changes the difficulty enormously.

28.6 Explain why Black's swaption formula is exact rather than approximate, being specific about what cancels.

28.7 Explain why a CMS coupon breaks the cancellation that makes Black's swaption formula exact, identify the numeraire its payoff calls for, and say what quantity the resulting adjustment consists of.

28.8 Explain why Girsanov changes drifts and leaves volatilities alone, and what this makes possible throughout the book.

28.9 Explain why $N(d_1)$ and $N(d_2)$ are probabilities of the same event, name the measure each is computed under, and say why they differ.

28.10 Derive Margrabe's formula by taking asset 2 as the numeraire, and explain why no interest rate appears in the result.

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28.11 State the two conditions a numeraire must satisfy, and give an instance in which each has failed in a real market.

28.12 Explain why the discount curve of Chapter 7, ColVA, and FVA are all statements about the choice of numeraire, and why that made the FVA debate unresolvable by data.

28.13 Using [TN28], state both fundamental theorems, and explain why stochastic volatility, jumps, and credit each produce incompleteness for a different specific reason.

FURTHER QUESTIONS

28.14 Derive the market-price-of-risk condition from the requirement that a riskless portfolio earn r, then show that risk-neutral valuation is the special case $\lambda = 0$ and explain in what sense the risk-neutral measure is a computational device rather than a belief.

28.15 Prove that volatility is invariant under a change of measure, and use the result to explain why a stochastic volatility model calibrated to option prices can be tested against realized volatility at all; then state precisely what such a test can and cannot detect.

28.16 The second fundamental theorem says incompleteness produces a range of arbitrage-free prices. For a stochastic volatility model, characterize that range, explain what determines where inside it the market trades, and connect your answer to Section 19.10's volatility risk premium.

28.17 This chapter is placed first in Part VI, on the argument that four of the part's six chapters would otherwise borrow machinery they have not been given. Construct the opposing case for teaching it last, on the argument that the theory is easier to motivate after seeing what it explains; identify which chapters would become harder and which easier, and give your own judgment with reasons.

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